HEAR-WORK AI: Public-Data Development and External Validation of an Explainable Framework for Occupational Stress, Tinnitus, and Hearing Outcomes

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Abstract

Purpose: This manuscript proposes HEAR-WORK AI, an American Journal of Audiology-targeted research framework for studying associations among occupational stress, tinnitus, and hearing outcomes using openly accessible population datasets and clinician-verified artificial intelligence (AI). The study is designed to avoid causal overclaiming: occupational stress is treated as a psychosocial exposure that may be associated with tinnitus burden, help-seeking, and hearing-related outcomes rather than as a proven direct cause of cochlear injury.

Method: We select the Korea National Health and Nutrition Examination Survey (KNHANES) as the primary development dataset and the U.S. National Health and Nutrition Examination Survey (NHANES) as the geographic external validation dataset. Both resources support population-level analyses of audiometry, self-reported tinnitus or hearing difficulties, noise exposure, and health covariates. We further specify an optional clinical extension using deidentified audiograms and otolaryngology records from participating hospitals. The AI component combines transparent statistical models, gradient-boosted trees, and an open medical language model such as MedGemma for schema-constrained extraction from deidentified clinical narratives. AI outputs are limited to feature extraction, calibrated risk estimates, and guideline-linked explanations.

Expected Contribution: HEAR-WORK AI aims to provide a reusable public-data baseline, an external-validation workflow, a clinician-review protocol, and open code for researchers studying tinnitus, hearing loss, workplace stress, and audiological outcomes. The framework prioritizes reproducibility, fairness, clinical safety, and global accessibility.

Conclusions: Public datasets can support robust association analyses and benchmark AI methods, but prospective clinical cohorts are required to evaluate treatment timing, recovery, and individual prognosis. HEAR-WORK AI offers a path from open epidemiological evidence to clinically grounded, explainable audiology AI.

Keywords: tinnitus; hearing loss; audiometry; occupational stress; KNHANES; NHANES; explainable AI; MedGemma; public datasets; audiology.

1 Introduction

Tinnitus and hearing loss are common auditory health concerns that influence communication, sleep, emotional well-being, work participation, and quality of life. In employed middle-aged adults, tinnitus may emerge in a context of psychosocial stress, long working hours, sleep disruption, occupational or recreational noise, cardiometabolic risk, and delayed help-seeking. These factors can be temporally and biologically intertwined, which makes simple causal narratives unreliable.

The HEAR-WORK AI study therefore begins from a conservative and testable premise: occupational stress may be associated with tinnitus and hearing outcomes, but public cross-sectional datasets are primarily suited for estimating associations rather than proving that stress directly causes tinnitus or sensorineural hearing loss. This distinction is particularly important for sudden or rapidly progressive hearing symptoms, which require timely clinical evaluation and should not be attributed to stress without audiological and otolaryngological assessment.

The practical goal is to support audiology and otolaryngology researchers through an open, reproducible framework. First, population datasets can identify robust associations and subgroups. Second, external validation across Korean and U.S. datasets can test whether findings generalize beyond one country. Third, open medical AI models can reduce the burden of structuring clinical narratives when prospectively collected hospital data become available. Fourth, an open repository can lower the barrier for global researchers to reproduce analyses, propose improvements, and share harmonized features.

This manuscript updates the prior HEAR-WORK AI protocol for the *American Journal of Audiology*. The revised emphasis is clinical audiology: pure-tone thresholds, tinnitus burden, speech-in-noise implications, unilateral hearing difficulties, hearing rehabilitation, and safe referral for urgent symptoms. AI is treated as an enabling tool, not as a replacement for clinicians.

2 Related Work

Population studies have repeatedly shown that tinnitus and hearing loss often coexist, but their relationships are heterogeneous. Analyses using KNHANES have estimated tinnitus prevalence and associated factors in Korea and demonstrated that nationally representative data can support clinically meaningful tinnitus research (Park et al., 2014; Joo et al., 2015). NHANES has similarly enabled U.S. studies of audiometry, occupational noise exposure, tinnitus, and mental health correlates (Hoffman et al., 2020; Chakrabarty et al., 2024; Mahboubi et al., 2013).

Recent audiology work highlights the need to measure more than pure-tone average. Executive-control studies in individuals with tinnitus suggest that cognitive burden and attentional control may be relevant to perceived functional impact (American Journal of Audiology, 2026a). Speech perception in noise and hearing-aid signal-management studies demonstrate that real-world listening difficulties can persist even when threshold-based metrics are incomplete (American Journal of Audiology, 2026b,c). These findings motivate HEAR-WORK AI to include tinnitus severity, communication difficulty, occupational participation, and listening effort as outcomes alongside audiometric thresholds.

Prior work involving Hun Yi Park and colleagues is especially relevant for the clinical expansion of this study. Sound-localization findings in unilateral hearing-aid users show that the side of hearing impairment may influence spatial-hearing interpretation (Ha et al., 2022). Early management research in traumatic benign paroxysmal positional vertigo, although not a tinnitus study, underscores the importance of rapid identification and pathway-based care for acute otologic and vestibular symptoms (Kim et al., 2022). Cochlear-implant research using a CT-derived basal turn–facial ridge angle illustrates how explainable anatomical variables can support prediction of residual hearing preservation (Kim et al., 2021). Together, these studies support a HEAR-WORK AI design that separates screening, referral, treatment, and rehabilitation endpoints.

Open medical AI models are increasingly available for research use. MedGemma is a collection of open models optimized for medical text and image comprehension and is positioned as a starting point for health-AI development rather than a validated diagnostic device (Google Research, 2025; Google Health AI, 2026). For HEAR-WORK AI, this means the model should be constrained to schema extraction and evidence-linked explanations, with clinician verification and external validation before any clinical deployment.

3 Public Dataset Selection

3.1 Primary Dataset: KNHANES

KNHANES is selected as the primary development dataset because it is nationally representative for Korea and has previously supported tinnitus and hearing-loss analyses. Its advantages for the present study include Korean population relevance, survey weights, audiometric data in applicable cycles, tinnitus and hearing-related questionnaire variables in selected cycles, health covariates, and psychosocial or stress-related self-report variables. It also provides a realistic bridge to future Korean hospital cohorts.

The limitations are equally important. KNHANES is not a workplace cohort, and many analyses are cross-sectional. Therefore, it can evaluate associations among perceived stress, work-related variables, tinnitus, and audiometric outcomes, but cannot by itself establish that occupational stress caused tinnitus or hearing loss. Treatment timing and recovery after sudden hearing loss are also not adequately captured.

3.2 External Validation Dataset: NHANES

NHANES is selected as the main external validation dataset because it provides openly downloadable U.S. public-use files, audiometry data in selected cycles, noise-exposure information, tinnitus or hearing-related questionnaires in selected cycles, and extensive demographic and health covariates. NHANES enables testing whether feature definitions and risk models derived from KNHANES have transportability across population, language, health system, and occupational contexts.

3.3 Optional Supplementary Datasets

Open audiology repositories such as the Oldenburg Hearing Health Repository, OpenNeuro tinnitus neuroimaging datasets, and Zenodo hearing-aid or auditory-test datasets may be used for secondary experiments. These datasets are not selected as primary because they typically lack the joint combination of population weights, tinnitus variables, audiometry, occupational factors, and general health covariates needed for the main association analyses.

4 Methods

4.1 Study Design

The proposed work has three stages. Stage 1 analyzes KNHANES as the primary public-data development cohort. Stage 2 evaluates transportability in NHANES. Stage 3 prepares a prospective clinical extension for participating otolaryngology and audiology clinics.

4.2 Participants and Harmonization

For KNHANES and NHANES, eligible records will include adults in the age range covered by audiometry modules, with emphasis on middle-aged employed participants when variables permit. The harmonization layer will map age, sex, education, occupation, employment status, perceived stress, sleep, smoking, alcohol, cardiometabolic disease, noise exposure, tinnitus status, hearing difficulty, and pure-tone thresholds into a common schema. Analysis code will preserve survey cycle, sampling strata, cluster identifiers, and weights.

4.3 Exposure and Outcomes

The primary exposure is stress-related self-report, operationalized according to each dataset's available questions. When direct occupational stress instruments are unavailable, the analysis will label the measure as perceived stress rather than occupational stress. Work-related variables will be treated as effect modifiers or covariates rather than as perfect substitutes for validated job-stress scales.

Primary outcomes are tinnitus status and audiometric hearing status. Hearing outcomes will include pure-tone average, frequency-specific thresholds, unilateral or asymmetric hearing patterns, and speech-frequency versus high-frequency impairment. Secondary outcomes include self-reported hearing difficulty, hearing-aid use where available, and mental-health or quality-of-life correlates.

4.4 Statistical Analysis

All public-data analyses will use survey-aware methods. Descriptive analyses will estimate weighted prevalence of tinnitus and hearing impairment. Association analyses will use weighted logistic, ordinal, or linear regression depending on outcome type. Models will be progressively adjusted for demographic factors, noise exposure, health behaviors, cardiometabolic factors, and mental-health covariates. Results will be reported as associations with confidence intervals, not causal effects.

Sensitivity analyses will test alternative hearing-loss definitions, age restrictions, exclusion of conductive patterns when tympanometry is available, and stratification by employment status, noise exposure, sex, and unilateral/asymmetric hearing status.

5 Open AI System and Research Acceleration

5.1 Motivation

A major barrier to audiology AI is the scarcity of harmonized, reusable clinical features. HEAR-WORK AI addresses this by publishing a transparent feature schema, reproducible public-data code, and a clinician-verification protocol. This supports a "benefit broadly shared" research model: investigators can reuse the pipeline, add local cohorts, and compare models without exposing identifiable patient data.

5.2 Model Roles

The AI system has four restricted roles. First, it harmonizes variable names and labels across public datasets. Second, it trains transparent baseline models for association and risk stratification. Third, in clinical extensions, an open medical language model such as MedGemma extracts structured fields from deidentified notes using a fixed schema. Fourth, it generates explanation drafts that cite only approved study documents or clinical guidelines.

The system is explicitly prohibited from diagnosing an individual patient, assigning a definitive cause, recommending steroids, or delaying urgent referral. Red-flag symptoms are handled by deterministic safety rules.

5.3 Robustness and Trustworthiness

Trustworthiness will be improved through seven design choices: (1) public-data baselines; (2) country-level external validation; (3) survey-weighted epidemiology; (4) interpretable models before complex models; (5) locked feature definitions; (6) clinician review of LLM-extracted variables; and (7) subgroup error analysis. Code will log dataset versions, variable mappings, model hyperparameters, prompts, and random seeds.

5.4 Open Science Contribution

The repository will include executable analysis templates, a data dictionary, model cards, prompt templates, and a reporting checklist. Because KNHANES files cannot be redistributed without following official access procedures, the code will provide directory expectations and variable mapping scripts rather than repackaging the raw data. NHANES download helpers will retrieve public files directly from CDC endpoints when available.

6 Planned Results

No numerical findings are reported in this pre-analysis manuscript. The planned results will include: weighted prevalence of tinnitus and hearing impairment in KNHANES; adjusted associations between perceived stress and tinnitus; adjusted associations between stress and audiometric thresholds; subgroup analyses among employed participants; NHANES external-validation performance; and model calibration, discrimination, and fairness metrics.

For transparency, all tables will distinguish observed public-data results from prospective clinical-extension results. Any LLM-extracted clinical fields will be reported with human verification accuracy, unsupported-output rate, and clinically harmful-output review.

7 Discussion

This AJA-targeted revision makes three changes to the earlier HEAR-WORK AI concept. First, it selects KNHANES as the most appropriate primary public dataset because it is Korean, population-based, and relevant to the intended clinical setting. Second, it uses NHANES as an external validation resource to test cross-national robustness. Third, it reframes AI as a reproducible research accelerator rather than a diagnostic authority.

The expected scientific contribution is not a claim that workplace stress directly damages hearing. Instead, HEAR-WORK AI aims to quantify whether stress-related measures are consistently associated with tinnitus, hearing difficulty, and audiometric outcomes after accounting for noise exposure and health covariates. This conservative framing improves credibility and aligns with what public datasets can support.

The clinical contribution is an audiology-centered pathway. Public data can identify population patterns and high-risk profiles; hospital data can later test treatment timing, sudden hearing-loss pathways, speech-in-noise outcomes, hearing-aid rehabilitation, and tinnitus burden. The open-source package is intended to help researchers in Korea, the United States, and elsewhere conduct comparable analyses using local cohorts.

Several limitations remain. Stress measurement differs across datasets and may not capture validated occupational stress constructs. Cross-sectional data cannot establish temporal order. Audiometry cycles vary across survey years. Tinnitus definitions and severity scales may differ. Public datasets usually lack detailed treatment timing. Finally, open medical language models require local validation, privacy review, and clinical governance.

8 Conclusion

HEAR-WORK AI provides a public-data-first and clinically grounded framework for studying tinnitus, hearing loss, and stress-related exposures. KNHANES is the preferred development dataset, NHANES is the preferred public external-validation dataset, and open medical AI models can accelerate feature extraction only under strict human verification. This design improves reproducibility, robustness, and global usefulness while avoiding unsupported causal claims.

Declarations

Author Contributions: Geunsik Lim conceptualized the HEAR-WORK AI framework, drafted the AI and open-science components, and prepared the reproducible repository structure. Hyun Jo contributed clinical audiology and otolaryngology framing and will guide future clinical validation planning. Both authors approved this draft.

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Data Availability: This manuscript uses publicly accessible dataset plans. KNHANES data are available through the Korea Disease Control and Prevention Agency subject to official access procedures. NHANES public-use files are available from the U.S. CDC. The GitHub repository will provide code, configuration files, and variable mapping templates, but it will not redistribute restricted raw data.

AI Use Disclosure: AI tools were used to structure the manuscript draft and code scaffold. In the proposed study, open medical language models will be used only for research feature extraction and explanation drafting from deidentified text, with clinician verification before analysis.

Conflicts of Interest: None declared in this draft.

Table 1: Selected public datasets for HEAR-WORK AI

Dataset	Role	Strengths	Main limitations
KNHANES	Primary development dataset	Korean population relevance; audiometry in selected cycles; tinnitus and health variables; survey weights	Cross-sectional; stress variables may be perceived stress rather than validated occupational stress; raw data access procedures required
NHANES	External validation dataset	Public CDC files; audiometry and noise exposure in selected cycles; rich health covariates; geographic validation	U.S. population; cycle-specific hearing and tinnitus variables; complex survey design
OHHR / Zenodo audiology data	Secondary method testing	Audiology-specific measurements; useful for model benchmarking	May lack stress, occupation, or tinnitus variables
OpenNeuro tinnitus datasets	Optional neurophysiology substudy	Public neuroimaging or fNIRS data for tinnitus biomarkers	Not designed for occupational stress or longitudinal treatment outcomes

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Table 2: AI components and safety constraints

Component	Intended use	Safety constraint
Survey harmonization	Map KNHANES/NHANES variables to common schema	Human-readable mapping file and version control
Statistical baseline	Estimate adjusted associations and transparent risk scores	Survey-aware methods; no causal claims from cross-sectional data
MedGemma or similar open medical LLM	Extract fixed fields from deidentified clinical notes in future cohorts	Clinician verification; no diagnosis or treatment recommendation
Retrieval-augmented explanation	Draft evidence-linked explanations for researchers and clinicians	Curated sources only; unsupported outputs flagged
Red-flag rule layer	Identify sudden hearing-loss or neurologic warning signs	Deterministic referral rules override probabilistic model

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